Assignment 2: X-ray Powder Diffraction (XRD) Diffractometer Simulation

Course: Nanotechnology & Quantum Physics | **Topic:** instrument simulation, Bragg-Brentano geometry, structure factors, crystallite size

Students: Mohamed Oboda (2111011090) | Mert SOLMAZ (2311011039)

Status: draft. The graded questions are not yet in this document because the assignment brief has not been received; everything below describes the simulator itself.

Item	What it is
Simulator	Cu K-alpha tube, divergence and Soller slits, flat sample on a theta/2theta goniometer, receiving slit, optional monochromator, detector with Poisson counting statistics
Phase database	Si, Al, Cu, alpha-Fe, NaCl, MgO, ZnO, TiO2 rutile and anatase, with the source of every cell constant in `data/SOURCES.md`
Physics	Bragg's law, structure factor with Waasmaier-Kirfel form factors, multiplicities, Lorentz-polarisation, Debye-Waller, K-alpha doublet, Caglioti resolution, Scherrer broadening, Poisson noise
Repository	https://github.com/OBODA0/quantum-physics

1. The instrument

The simulator models the chain in Figure 1. A copper tube emits the K-alpha doublet at 1.540598 and 1.544426 angstrom, with K-beta removed by a filter or a monochromator. The beam leaves a divergence slit, is collimated by Soller slits, strikes a flat powder specimen on a theta/2theta goniometer, and reaches the detector through a receiving slit. The measured variable is 2theta; the specimen sits at theta, so a reflection appears when 2 d sin(theta) = n lambda.

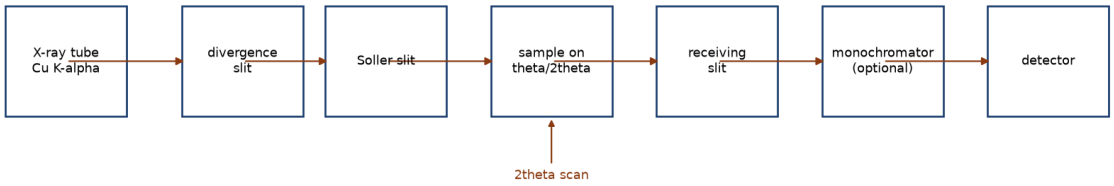


Figure 1: the chain the simulator models. Every box is a module: the tube and lines in `sources.py`*, the slits and flat-plate geometry in* `geometry.py`*, the specimen in* `samples.py`*, the detector in* `detector.py`*, and the scan in* `diffractometer.py`*.*

2. The model

Positions come from Bragg's law and the cell. Intensities are relative, as in assignment 1: the model reproduces patterns and intensity ratios, not absolute radiometry. For each reflection the relative intensity is |F|^2 m LP, where m is the powder multiplicity and LP = (1 + cos^2 2theta) / (sin^2 theta cos theta) without a monochromator. The structure factor sums the basis with the Waasmaier-Kirfel form factor f0(s), s = sin(theta)/lambda, and a Debye-Waller term exp(-B s^2).

Each line is drawn with a pseudo-Voigt profile whose FWHM follows the Caglioti form sqrt(U tan^2 theta + V tan theta + W), combined with the Scherrer width K lambda / (D cos theta) for a crystallite size D. Counts are Poisson draws around the expected mean; the detector's dead time is available as a control.

The model's absences are not tabulated. The 222 line of diamond, the 100 and 110 of an FCC metal and the 100 and 111 of a BCC metal are zero because the summed phases cancel, which is a check that the basis is right.

3. The phase database

Key	Phase	Space group	Cell (A)	Strongest line
si	Silicon (Si)	Fd-3m (227)	a = 5.4311	111 at 28.44 deg
al	Aluminium (Al)	Fm-3m (225)	a = 4.0495	111 at 38.47 deg
cu	Copper (Cu)	Fm-3m (225)	a = 3.6149	111 at 43.32 deg
fe	alpha-Iron (Fe)	Im-3m (229)	a = 2.8665	110 at 44.67 deg
nacl	Rock salt (NaCl)	Fm-3m (225)	a = 5.6402	200 at 31.70 deg
mgo	Periclase (MgO)	Fm-3m (225)	a = 4.2112	200 at 42.92 deg
zno	Zincite (ZnO)	P63mc (186)	a = 3.2494, c = 5.2038	101 at 36.26 deg
tio2-rutile	Rutile (TiO2)	P42/mnm (136)	a = 4.5937, c = 2.9587	110 at 27.44 deg
tio2-anatase	Anatase (TiO2)	I41/amd (141)	a = 3.7892, c = 9.5370	101 at 25.27 deg

Table 1: the phases the simulator ships. Cell constants, their sources and their confidence are listed in `data/SOURCES.md`*; every row is checked against its published strongest reflection in the test suite.*

4. Results

Figure 2 shows the nine phases under the same instrument settings. The differences are the physics: an FCC metal has few strong lines, a rocksalt has a weak 111 because the two form factors nearly cancel, wurtzite breaks the cubic degeneracy, and the two titania polymorphs give different patterns from the same formula.

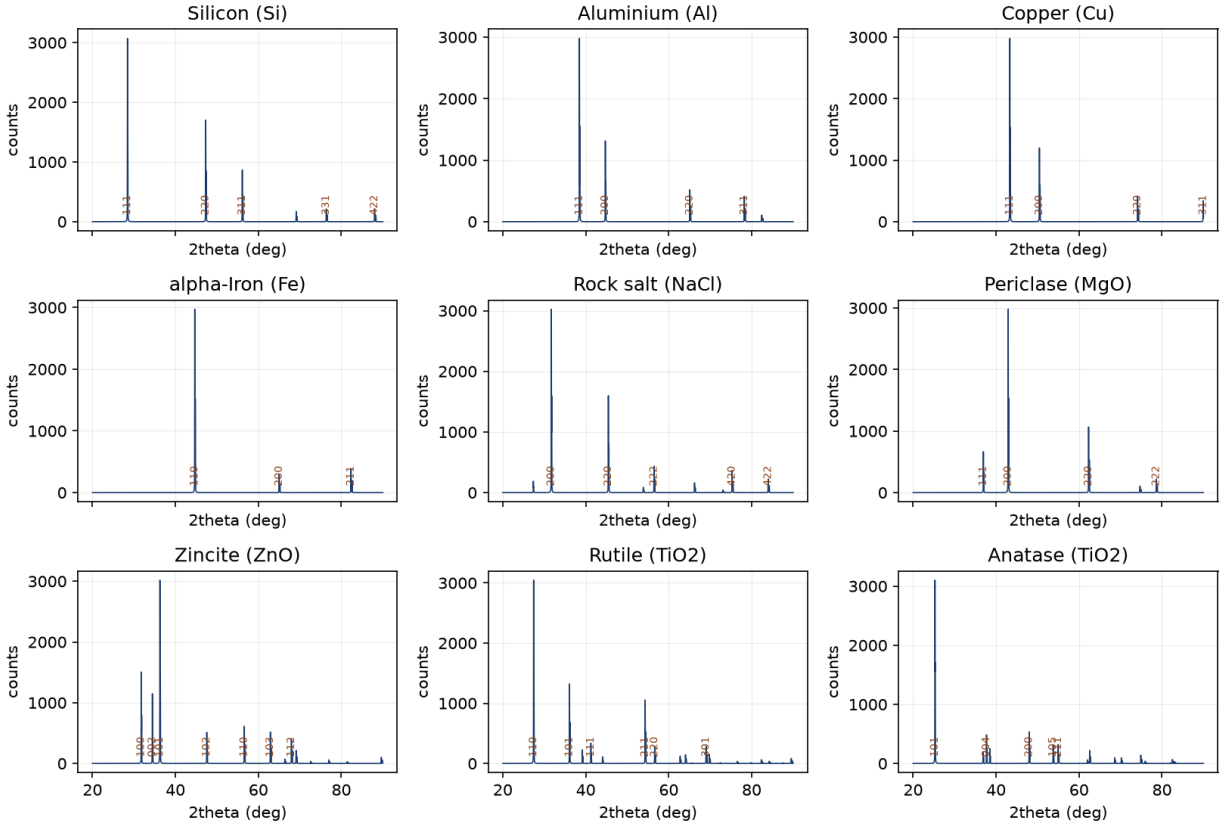


Figure 2: simulated patterns, 20-90 deg, Cu K-alpha, 20 000 counts at the strongest line, no noise. The hkl labels are the reflections the model predicts.

4.1 Crystallite size

Broadening is the sum of the instrument and the specimen. Figure 3 fixes the instrument (U, V, W) and varies the crystallite size; the Scherrer analysis in `xrdsim/analysis.py` subtracts the instrumental width in quadrature and recovers the input size to within about 1 % on a noiseless scan (30.0-30.1 nm for an input of 30 nm).

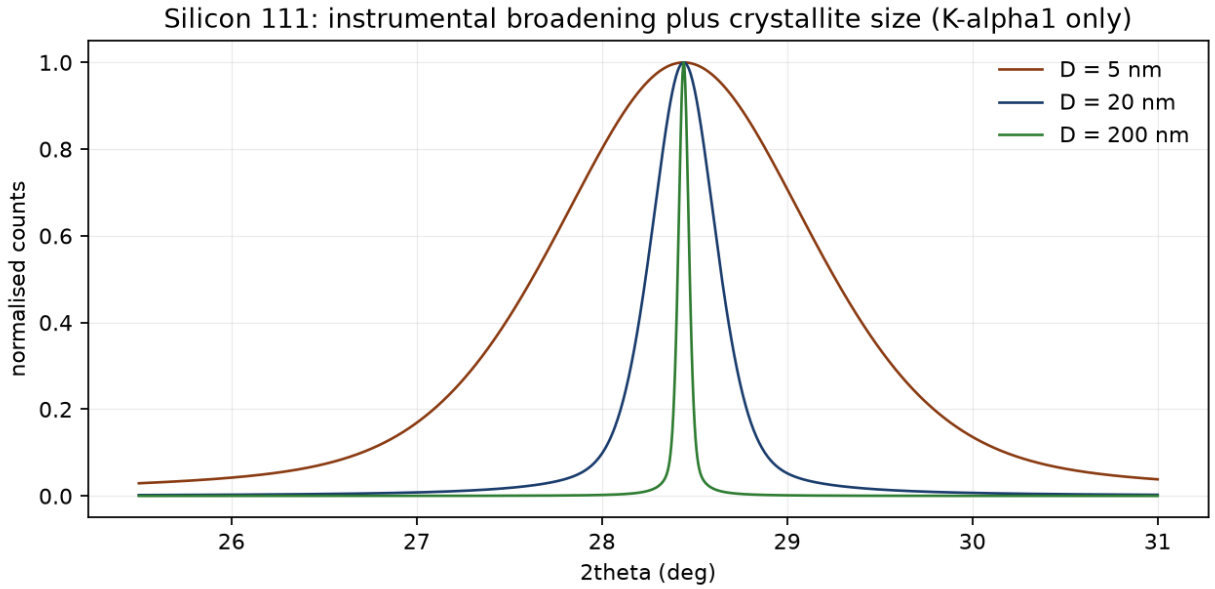


Figure 3: the silicon 111 line at 5, 20 and 200 nm. The curves are normalised; the K-alpha2 line is switched off so that only the intrinsic width remains.

4.2 The K-alpha doublet

The doublet separation grows with angle, Delta(2theta) = 2 tan(theta) (lambda2 - lambda1)/lambda1: about 0.07 deg at 28 deg and 0.28 deg at 88 deg. Below 60 deg the two lines merge into one asymmetric peak, which is why a fitted width at low angle is not the instrument width.

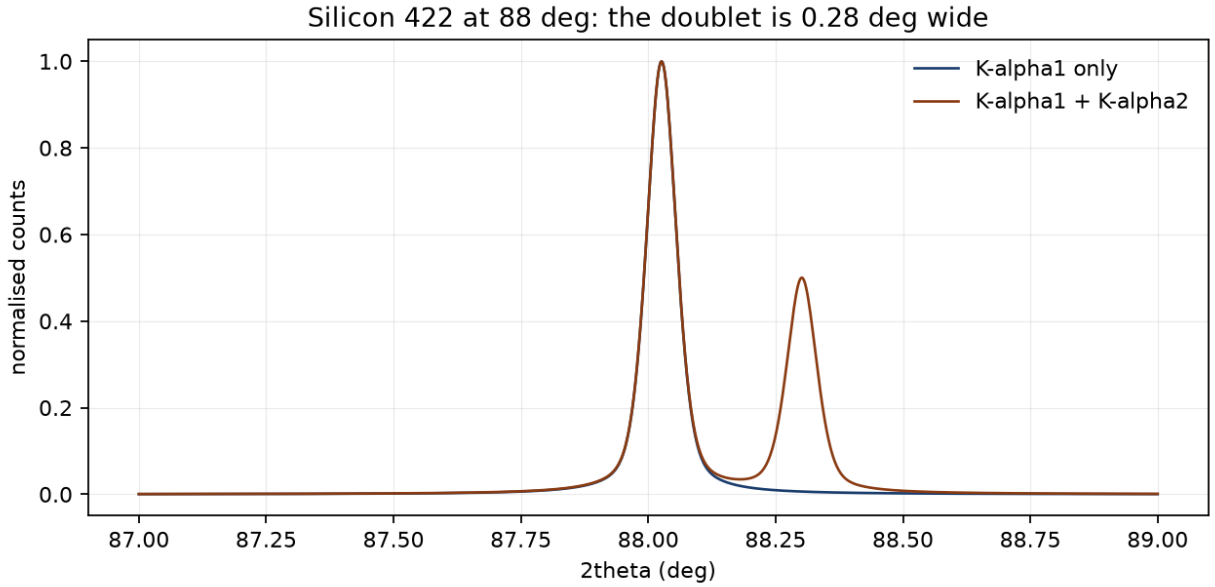


Figure 4: silicon 422 with and without the K-alpha2 line, at the same instrument settings.

5. Verification

The suite covers lattice geometry, profiles, the detector, the structure factor and its absences, the scan, the analysis and the command line interface. Independent checks compare simulated peak positions with certified and published values: silicon 111/220/311/400 at 28.441/47.301/56.120/69.127 deg against the NIST SRM 640d values 28.4425/47.3034/56.1211/69.1295, and the relative intensities 100/58/30/6 against the standard 100/55/30/5. The 0.002-0.007 deg residual is the wavelength convention: the certificates quote 1.5405929 angstrom, the simulator uses 1.540598.

Multiplicities are checked against a brute-force enumeration of the symmetry operations for cubic, tetragonal and hexagonal families, and the intensity ratios sit inside the spread expected from a uniform Debye-Waller factor and no absorption correction: copper 111/200/220 comes out at 100/41/16 against the published 100/46/20, and rutile at 100/43/37 for 110/101/211 against 100/50/60. The whole assignment rebuilds from a fresh clone: uv sync, 91 tests, figures, tables, PDF and the gate.

Defects found and fixed during the build are listed in `VERIFICATION.md`: a missing step factor in the profile, duplicate reflections from permuted indices, a Lorentz-polarisation divergence at 180 deg, cell constants taken from primitive-cell conversions, and a hand-written anatase basis that produced a forbidden 001 line.

6. What is not modelled

Rietveld refinement, anomalous dispersion, texture beyond a single orientation factor, K-beta residual, detector energy discrimination and absolute intensities. The Cu K-alpha wavelengths are provisional: no primary table could be read from the build host. No simulated pattern has yet been compared with a measured scan; the comparisons above are with published peak positions and intensity ratios.